Assignment 2
Network Scanning & Threat Analysis

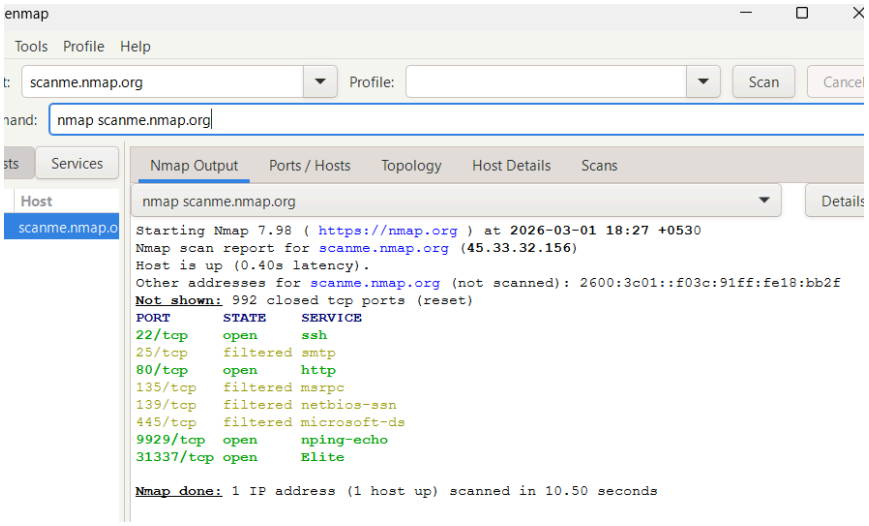
1. Nmap Commands Used:

-> Basic scan command:

nmap scanme.nmap.org

Explanation:

- Shows open ports
- Used to identify basic network exposure

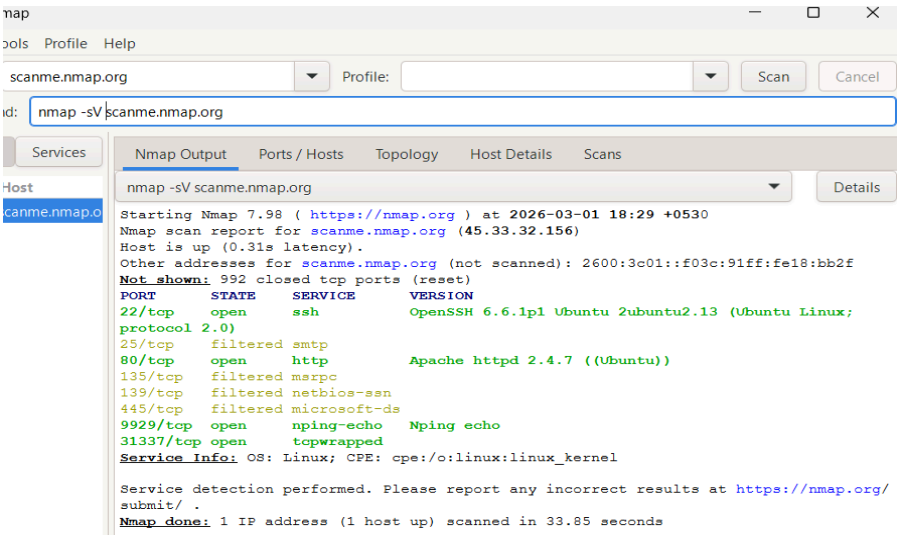


-> Service Version Detection:

nmap -sV scanme.nmap.org

Explanation:

- -sV → Detects service and software version

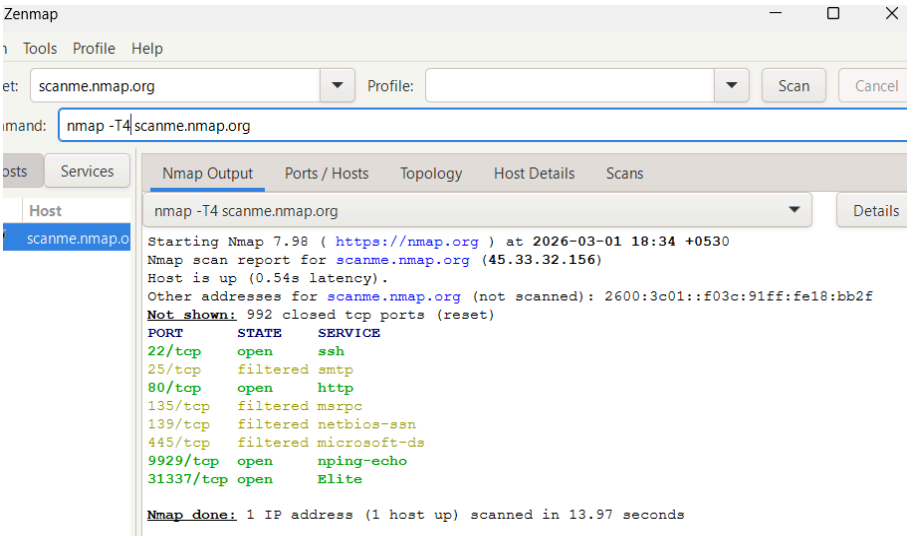


-> Fast Scan with Timing Template:

nmap -T4 scanme.nmap.org

Explanation:

- **-T4 → Sets aggressive timing.**
- **Increases scan speed.**
- **Open ports quickly.**

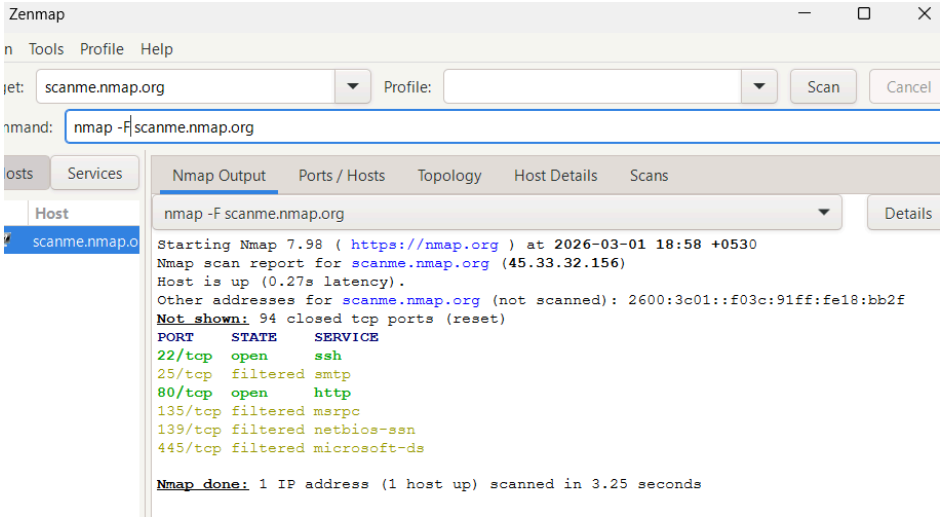


-> Fast Port Scan:

nmap -F scanme.nmap.org

Explanation:

- **-F → Fast scan.**
- **Quick overview of exposed services.**



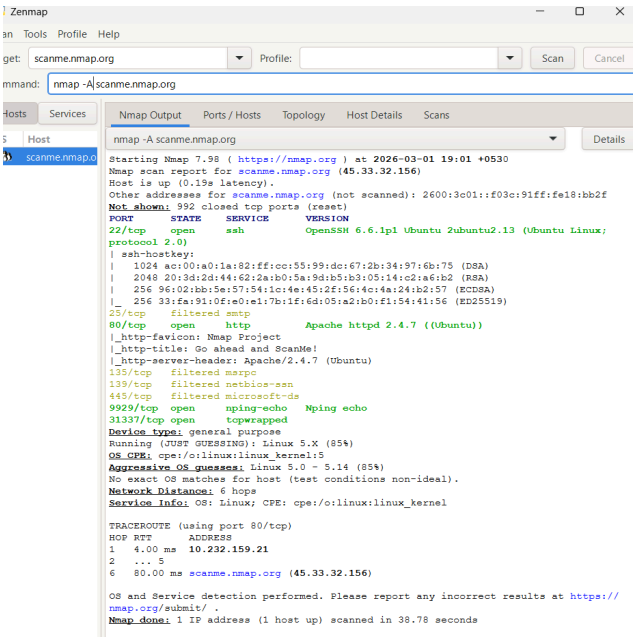
->Aggressive Scan:

nmap -A scanme.nmap.org

Explanation:

The -A option enables:

- **OS detection**
- **Version detection**

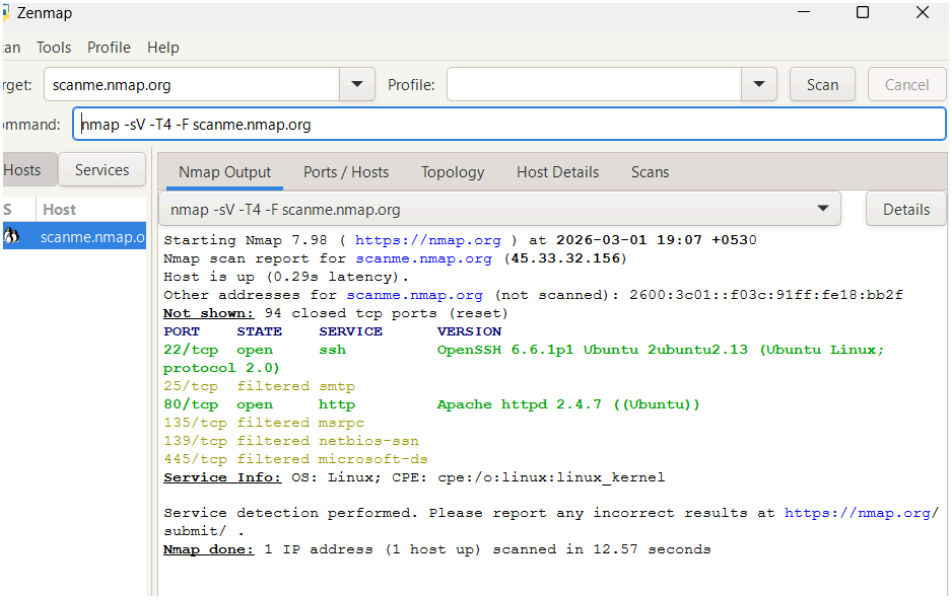


->Combined Command Used:

nmap -T4 -F -sV scanme.nmap.org

Explanation:

- **Fast scan (-F)**
- **Faster timing (-T4)**
- **Service and version detection (-sV)**



->XML Output for Analysis:

nmap -sV -oX scan.xml scanme.nmap.org

Explanation:

- **Saves scan results in XML format.**
- **Used for automated analysis using Python.**

2. Open Ports table:

Port	Protocol	State	Service	version/product
22	TCP	Open	SSH	OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13
80	TCP	Open	HTTP	Apache httpd 2.4.7 (Ubuntu)
9929	TCP	Open	nping-echo	Nping echo service
31337	TCP	Open	tcpwrapped	Access controlled

3. Virustotal finding, port & service analysis, vulnerability reasoning:

***************notebook attached**************

nmap_url_ip.ipynb

VirusTotal Findings:

VirusTotal was used to collect passive intelligence about the domain.

Purpose of VirusTotal:

VirusTotal aggregates results from multiple security engines and provides:

- **Reputation analysis**
- **Historical intelligence**
- **Threat detection information**

Passive vs Active Intelligence:

Active Scanning(Nmap)	**Passive Intelligence(virusTotal)**
Sends packets to target	Uses existing security data
Finds open ports/services	Finds reputation/history
Generates network traffic	No direct interaction

Port & Service Analysis:

Port 22 – SSH

- **Used for secure remote login.**

Port 80 – HTTP

- **Web server access.**

Port 9929 – Nping Echo

- **Used for Nmap testing.**

Conclusion:

This exercise demonstrated the process of reconnaissance and vulnerability analysis.

Key Learnings:

- How Nmap identifies open ports and services.
- Importance of different scan options for deeper information.
- Difference between active scanning and passive intelligence.
- How VirusTotal complements Nmap findings.
- How exposed services can indicate potential security risks.